

Max-Flow-Based OCS Reconfiguration Scheduling for Full-Stack Distributed LLM Training Acceleration in Optical AIDC

Dianxuan Fu, Xingyu Liu, Qile Liu, Jiaqiu Li, Yihao Zhang, Qizhi Qiu, Xiaomin Liu, Weisheng Hu, and Qunbi Zhuge*

State Key Laboratory of Photonics and Communications, School of Information Science and Electronic Engineering, Shanghai Jiao Tong University, Shanghai, 200240, China, *qunbi.zhuge@sjtu.edu.cn

Abstract We propose a max-flow-based OCS reconfiguration scheduling framework on our full-stack scale-out AIDC platform to accelerate LLM training. The proposed method reduces iteration time by up to 65.58% and improves All-Gather flow completion time by up to 15.36 $\times$ under heuristically pruned GPU grouping. ©2026 The Author(s)

Introduction

The explosive growth of large language models (LLMs) has sharply increased the demand for more computing resources within artificial intelligence data centers (AIDCs), pushing AI infrastructure towards its power and energy limits [1]. As Moore's Law decelerates, the distributed nature of LLM training necessitates massive data exchange among tens of thousands of graphics processing units (GPUs), making communication a major bottleneck during iterative training [2]. Therefore, optical circuit switching (OCS) technology has emerged as an inherently compatible scale-out solution for next-generation AIDCs [3-4], driven by its superior energy efficiency and high-bandwidth characteristics [4-5].

Despite these advantages, efficiently operating OCS in LLM training remains extremely challenging. OCS devices are bufferless and incur non-negligible reconfiguration delays [6]. During each reconfiguration event, optical connectivity is temporarily interrupted, forcing in-flight traffic to be absorbed by electrical packet switches (EPS). Since OCS provides no buffering, prolonged reconfiguration can quickly overflow EPS buffers, causing transient congestion and packet deflection [3-6]. Even small mismatches between traffic demand and circuit availability may cause re-transmissions or idle periods [5]. These penalties accumulate over iterations and significantly slow training. Furthermore, the coexistence of data parallelism (DP), tensor parallelism (TP), and

pipeline parallelism (PP) produces complicated traffic patterns [7-8]. This complexity makes it more difficult to coordinate OCS reconfiguration with full-cycle LLM training traffic, which is large-scale, time-varying, and overlaps with computing. Yet, this problem remains insufficiently explored in prior works on complex LLM workloads [9-11].

In this work, a max-flow-based OCS reconfiguration circuit scheduling and routing strategy is proposed to accelerate LLM training in optical AIDCs. We develop a full-stack optical AIDC simulation platform to support realistic large-scale LLM workload generation and OCS scheduling optimization. A lightweight heuristic mapping step determines an optimized parallelism and GPU grouping before training. Building on this foundation, we formulate OCS-mediated collective traffic as a multi-source multi-sink max-flow problem [12] and apply Dinic's algorithm [13] to drive demand-aware OCS reconfiguration and routing. Results show that the max-flow scheduling framework decreases iteration time by up to 65.58% and delivers an up to 15.36-fold improvement in collective flow completion time (FCT).

Principles

Fig. 1 illustrates the workflow of our full-stack optical AIDC simulation platform and max-flow-based OCS scheduling framework for LLM training acceleration. First, the AI workload is generated from the given LLM and AIDC configurations

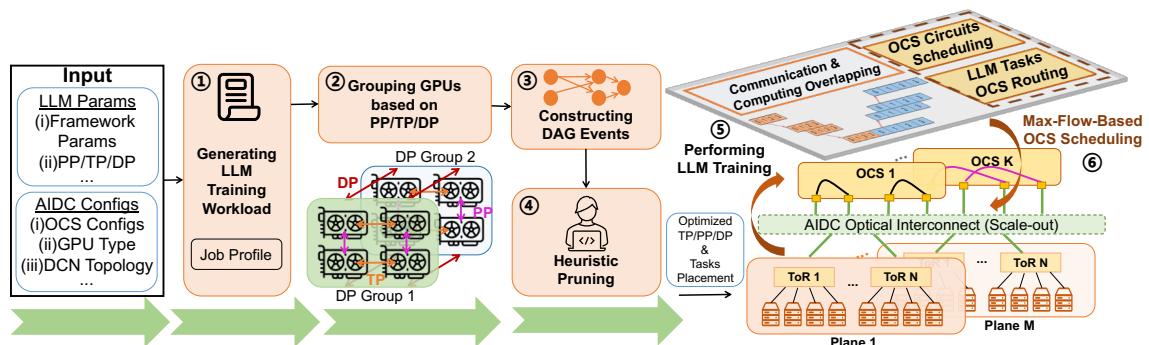


Fig. 1: The workflow of our full-stack optical AIDC simulation platform with max-flow-based OCS scheduling for training.

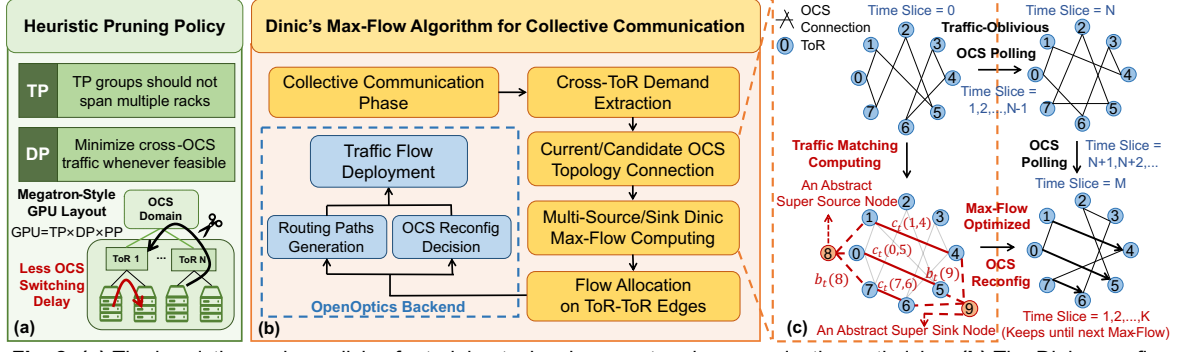


Fig. 2: (a) The heuristic pruning policies for training tasks placement and communication optimizing. (b) The Dinic max-flow algorithm flowchart for cross-ToR collective communication. (c) Schematic diagram of OCS switching by time slice.

(step ①). Based on the resulting job profiles, GPUs are then mapped into DP, TP and PP groups for collective communication emulation (step ②). The workload is next abstracted into directed acyclic graphs (DAGs) based on computing and communication events of each parallelism group (step ③). The DAGs capture both the sequential constraints of iterative training and the potential for communication-computation overlap. In step ④ (detailed in Fig. 2(a)), heuristic policies are applied to prune the large search space of parallelism combinations and GPU layouts. Eventually, the training process is executed according to the previous DAG and the optimized tasks placement results (step ⑤), together with the continuous scheduling of the max-flow algorithm (step ⑥, detailed in Fig. 2(b)-(c)). We next describe the details of step ④ and step ⑥.

Heuristic Pruning Policies (step ④, Fig. 2(a)): We apply heuristics to reduce the search space of parallelism configurations and GPU groups. Since collective communication such as All-Reduce is often bursty [2], communication-intensive traffic should remain within high-bandwidth domains. TP is particularly communication-intensive because each layer is partitioned across multiple GPUs and requires frequent intra-layer synchronization [8]. We therefore prioritize placing TP groups within a single rack and constrain the TP degree within the number of GPUs available in one rack (**Heuristic Policy 1**). Moreover, DP synchronizes gradients across replicated groups [7]. Although less frequent than TP communication, DP traffic often carries larger volumes, so we minimize DP traffic across OCS domains whenever possible (**Heuristic Policy 2**).

Dinic's Max-Flow Algorithm for Collective Communication (step ⑥, Fig. 2(b)): During the process of training, the max-flow algorithm for the cross-OCS traffic in collective communication phase is applied. We first extract the traffic that crosses Top-of-Rack switches (ToRs) and construct a ToR-level graph $G_t = (V_t, E_t)$ for time slice t , as illustrated in Fig. 2(c). Because collective communication generates concurrent many-

to-many transfers rather than a single point-to-point flow, the original problem is formulated as a multi-source multi-sink flow allocation problem. Let $b_t(v)$ denote the traffic demand injected from or required by ToR v . As shown in Fig. 2(c), an abstract super source s^* and a super sink d^* are introduced. The graph is converted into $G_t' = (V_t \cup \{s^*, d^*\}, E_t')$. For each source ToR u , we add an edge (s^*, u) with capacity $b_t(s^*)$. For each destination ToR v , we add an edge (v, d^*) with capacity $b_t(d^*)$. The capacity of each optical link edge is defined as:

$$c_t(u, v) = x_t(u, v)B(u, v)\Delta_t^{\text{eff}} \quad (1)$$

where $x_t(u, v) \in \{0, 1\}$ indicates the availability of ToRs u and v in time slice t , $B(u, v)$ is the bandwidth, and Δ_t^{eff} is the effective transmission window. Therefore, the max-flow problem is expressed as:

$$\max \sum_{(s^*, u) \in E_t'} f_t(s^*, u), \quad (2)$$

subject to

$$0 \leq f_t(i, j) \leq c_t(i, j), \forall (i, j) \in E_t', \quad (3)$$

$$\sum_{(k, i) \in E_t'} f_t(k, i) = \sum_{(i, j) \in E_t'} f_t(i, j), \forall i \in V_t. \quad (4)$$

where the edge capacity $c_t(i, j)$ is modelled as the effective traffic volume that can be delivered over an optical link during time slice t . We solve this transformed graph using Dinic's algorithm, which iteratively builds a level graph by breadth-first search (BFS) and computes blocking flows by depth-first augmentation until no further (s^*, d^*) path exists. As illustrated in Fig. 2(c), unlike traffic-oblivious polling, the proposed max-flow scheme establishes circuits according to the extracted traffic and preserves the optimized topology across subsequent slices until the next demand update and max-flow optimization. The resulting flow allocation ranks ToR-pair demand on candidate OCS links, allowing the scheduler to activate high-load circuits connections and generate corresponding routing paths, as shown in Fig. 2(b).

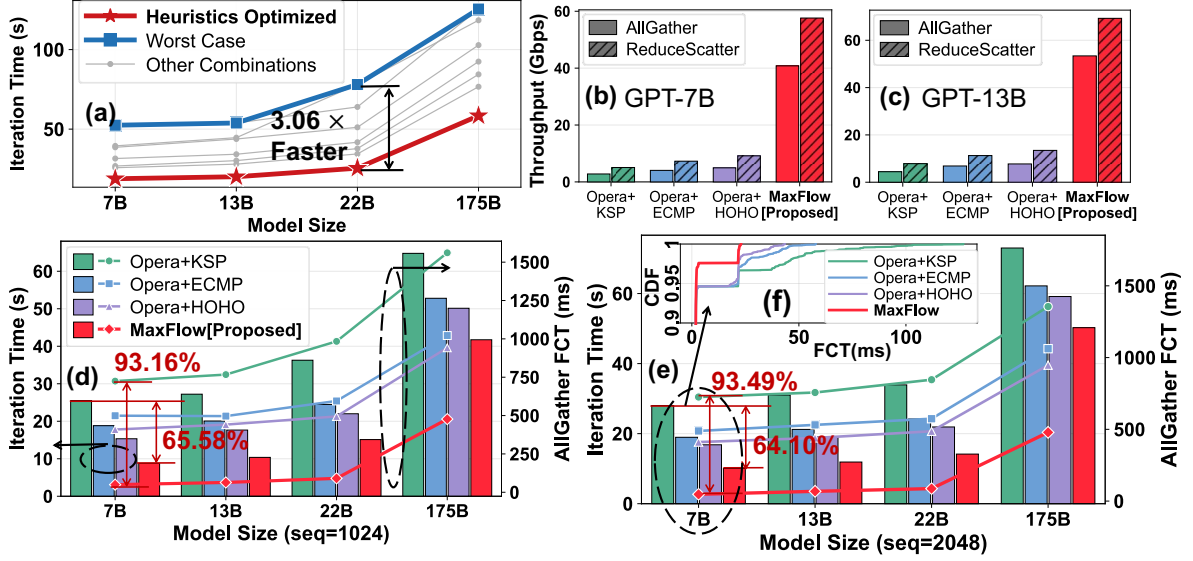


Fig. 3: (a) Heuristic pruning results. Throughput results for (b) GPT-7B and (c) 13B. The Iteration time across different model sizes with sequence lengths of (d) 1024 and (e) 2048. (f) Single-flow FCT CDF results of GPT-7B at 2048 sequence length.

Evaluation Setups and Results

The proposed method is evaluated on our full-stack optical AIDC simulation platform. The LLM workloads are generated using SimAI [14]. Training jobs are emulated in Megatron framework [7] with different combinations of TP, PP, and DP on an NVIDIA H200 GPU. We consider 4 model scales: GPT-7B, 13B, 22B, and 175B. The network is implemented in OpenOptics [15] as a 64-GPU optical AIDC with 8 servers, 8 ToRs, and 4 parallel OCS planes (as shown in Fig. 1, $M=K=4$, $N=8$). Each server hosts 8 GPUs and connects to a dedicated ToR with 100 Gbps links. Each ToR further connects to 4 independent 100 Gbps OCS uplinks. We utilize OpenOptics to evaluate both the Opera [16] baseline and the max-flow-based method. The OCS switching time slice is set to 20 ms. Link delay is modelled utilizing alpha-beta model [17] and calendar queue [15]. For routing strategy comparison, k-shortest path (KSP) [18], equal cost multi-path (ECMP) [19], and hop-on-hop-off (HOHO) [20] are implemented.

We first validate the impact of the heuristic GPU grouping policies. As shown in Fig. 3(a), under Opera scheduling and ECMP routing, the optimized layout consistently outperforms other parallelism combinations across all model sizes (sequence length=1024), reducing iteration time by up to 67.39% and reaching a 3.06 $\times$ speedup over the worst case for 22B. The selected combination, $TP=4$, $PP=8$, $DP=2$, follows our heuristic principles and effectively narrows the search space for subsequent OCS scheduling.

Fig. 3(b) and Fig. 3(c) compare the throughput for GPT-7B and GPT-13B at a sequence length of 1024 under different routing and scheduling strategies. The proposed max-flow method consistently achieves the highest throughput for both All-Gather and Reduce-Scatter. Fig. 3(d)

and Fig. 3(e) further evaluate iteration time under sequence lengths of 1024 and 2048, respectively. Across all tested LLM sizes, the proposed method consistently delivers the lowest iteration time. For sequence length 1024, it reduces iteration time by up to 65.58% and lowers All-Gather FCT by up to 93.16%. Under sequence length 2048, the corresponding gains reach 64.10% and 93.49%. Fig. 3(f) further shows that the proposed method yields a steeper single-flow FCT cumulative distribution function (CDF) with a much shorter tail than the baselines. These results confirm that the proposed scheduler adapts effectively to complex workloads and reduces both iteration time and flow-level latency.

The observed performance gains stem from LLM-demand-aware OCS scheduling. Opera rotates OCS connections on a fixed schedule. In contrast, the proposed max-flow-based method explicitly extracts cross-ToR demands, computes a global max-flow allocation, and prioritizes high-demand ToR pairs during reconfiguration.

Conclusions

In this work, we propose a max-flow-based OCS scheduling framework to accelerate LLM training in optical AIDCs. We construct a full-stack scale-out optical AIDC simulation platform for end-to-end LLM training and OCS scheduling evaluation. A lightweight heuristic pruning step is introduced to reduce the search space of parallelism configurations. Subsequently, we formulate OCS-mediated collective traffic as a max-flow problem and apply Dinic's algorithm to guide OCS reconfiguration and routing-path selection. Results show that the heuristic optimization achieves an up to 3.06 $\times$ acceleration in iteration time, while the max-flow-based traffic-matching framework reduces iteration time by up to 65.58% and improves All-Gather FCT by up to 15.36 $\times$.

Acknowledgements

This work was supported by Shanghai Pilot Program for Basic Research - Shanghai Jiao Tong University (21TQ1400213) and Open Project in 2025 (Project2: Research on Mechanism of Ultra-High-Speed Large-Capacity All-Optical Switching for Satellite Network) from State Key Laboratory of Satellite Network and Shanghai Key Laboratory of Satellite Network.

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